

HPS-GPR v5.5.1

Testing the energy-response option across structures and dataset pairs

11 September 2026 | Extension of the v5.5.0 diagnostic

What survives the wider comparison

The additional response $R(E) \propto E^{-2.904}$ remains a useful description of the selected 90–94 MeV rate pattern. It does not describe all other candidate structures consistently. The 66 MeV three-dataset fit prefers a much shallower slope, and several 76–79 MeV fits select extreme positive slopes by suppressing datasets with deficits. The 2016+2021 structures at 121 and 160 MeV prefer slopes near -4.65 , with broad profile shapes.

The combinatorial pair study evaluates every supported pair at each selected mass. Two positive rates can be reproduced exactly by an amplitude and a slope, so good pair fits alone are automatic. Transfer of the original fixed slope, slope stability, and checks against the unused dataset carry more information.

Model and comparisons

At each fixed mass, the inherited count model is

$$\lambda_i = b_i + L_i \theta_i + \mu K_i(m) (E_i / 2.30 \text{ GeV})^\beta, \quad \mu \geq 0.$$

The energies are 1.056, 2.30 and 3.74 GeV for 2015, 2016 and 2021 [1]. K_i gives counts per unit ϵ^2 using the existing prompt-density and effective-radiative-fraction conversion. The extra factor is a residual response *after* that normalization, not an additional theoretical production factor [2, 3]. Beam energy is confounded with detector, target and selection changes.

Compare $\beta = 0$, the frozen v5.5.0 value $\beta_{92} = -2.90358457$, and a newly fitted β at each mass. In each comparison the common nonnegative amplitude is refitted. Define

$$D = 2 \left[\text{NLL}_{\text{model}} - \sum_i \text{NLL}_{i, \mu_i \geq 0} \right].$$

Each D compares the same local data and constraints. The independent comparator also imposes nonnegative amplitudes; signed fits remain diagnostic displays. No likelihoods from different mass windows or overlapping dataset subsets are added.

Selection and scope

The saved half-MeV scans supply interior local maxima with signed $r \geq 1$ in each pair and the all-three scope. Within each scope, candidates are ranked by r and separated by at least twice the larger maximum active resolution of the two masses. This retains 14 scope-specific peaks, or 12 distinct masses. Adding inherited controls gives 17 masses, 56 scope–mass comparisons, 47 standalone fits and 39 unused-dataset checks. Selection precedes the new slope fits but remains retrospective; a selected combined maximum need not have positive rates in every constituent dataset.

All slopes are examined on $[-12, 12]$, with the old $[-6, 6]$ range retained for comparison. Full grids, local minima and endpoints are checked. Connected $\Delta(2\text{NLL}) \leq 1$ spans summarize profile shape; they are not calibrated confidence intervals. No new toys or significance calibration are introduced.

Does the 92 MeV slope transfer?

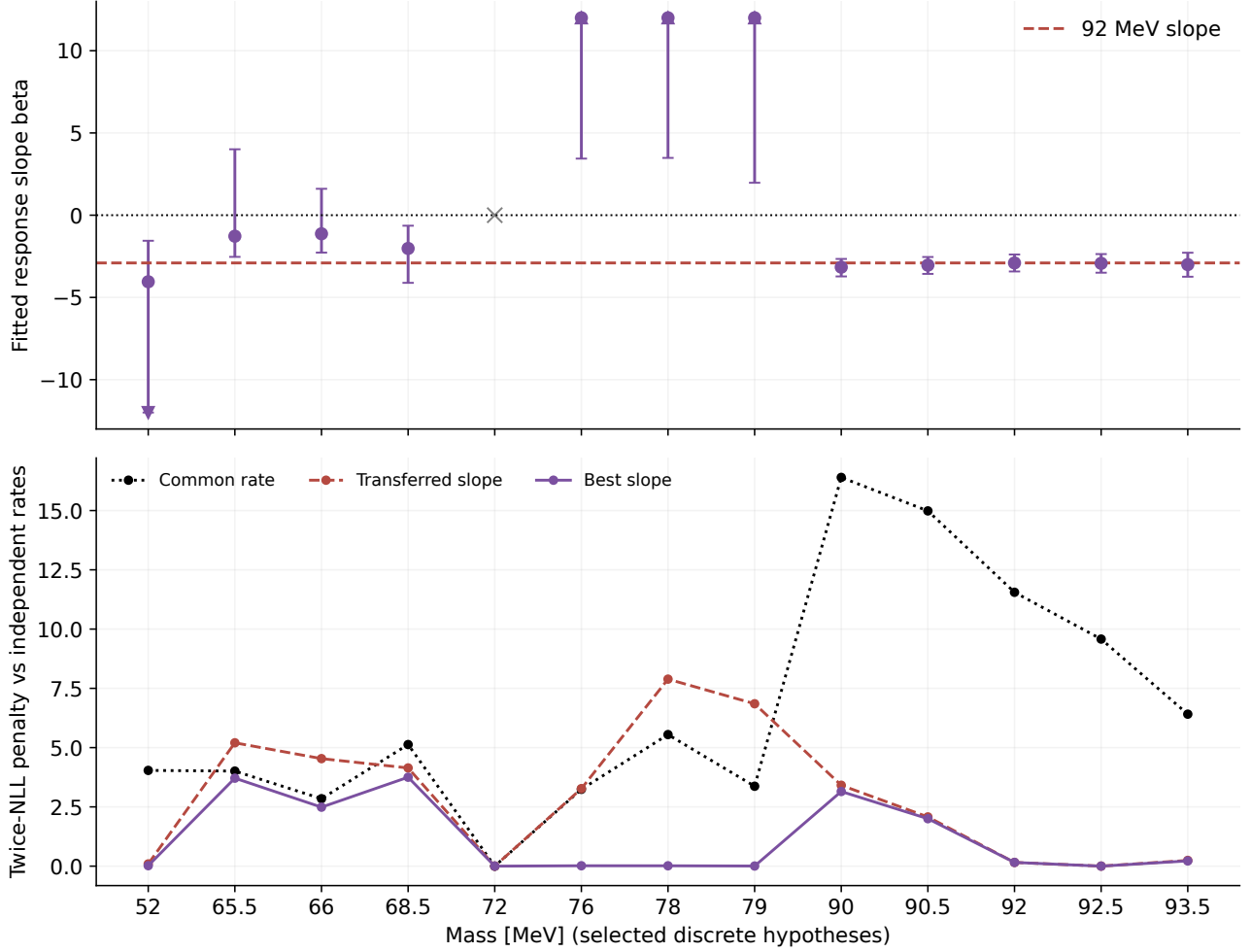


Figure 1: All supported three-dataset comparisons. Top: best slopes and connected unit-deviance profile spans; triangles mark ranges truncated by the search bounds. The gray cross at 72 MeV marks zero common amplitude and an unidentified slope. Bottom: penalties relative to independent nonnegative rates. Horizontal positions are selected hypotheses, not a uniformly spaced mass scan; connecting lines guide the eye.

At 92 MeV, the slope is -2.904 , with a profile span $[-3.422, -2.391]$. Its penalty is 0.157, versus 11.552 for a common rate. The 90.5, 92.5 and 93.5 MeV best slopes are -3.038 , -2.927 and -3.004 . These nearby fits share events and background constraints: their agreement is local stability, not independent confirmation. The near-zero penalty at 92.5 MeV was not used to select the mass or to replace the 92 MeV anchor.

At 66 MeV, transferring the frozen slope gives $D = 4.536$, compared with 2.845 for a common rate and 2.487 at the best slope -1.130 . At 78 MeV, the frozen penalty is 7.888. The apparent improvement to 0.015 at $\beta = 12$ comes from assigning almost all positive signal to 2021, while the other signed amplitudes are negative. Expanding the bound from 6 to 12 changes that slope with only a 0.270 reduction in twice NLL. This is an unstable suppression limit, not a measured steep production law.

The fitted rate patterns at representative masses

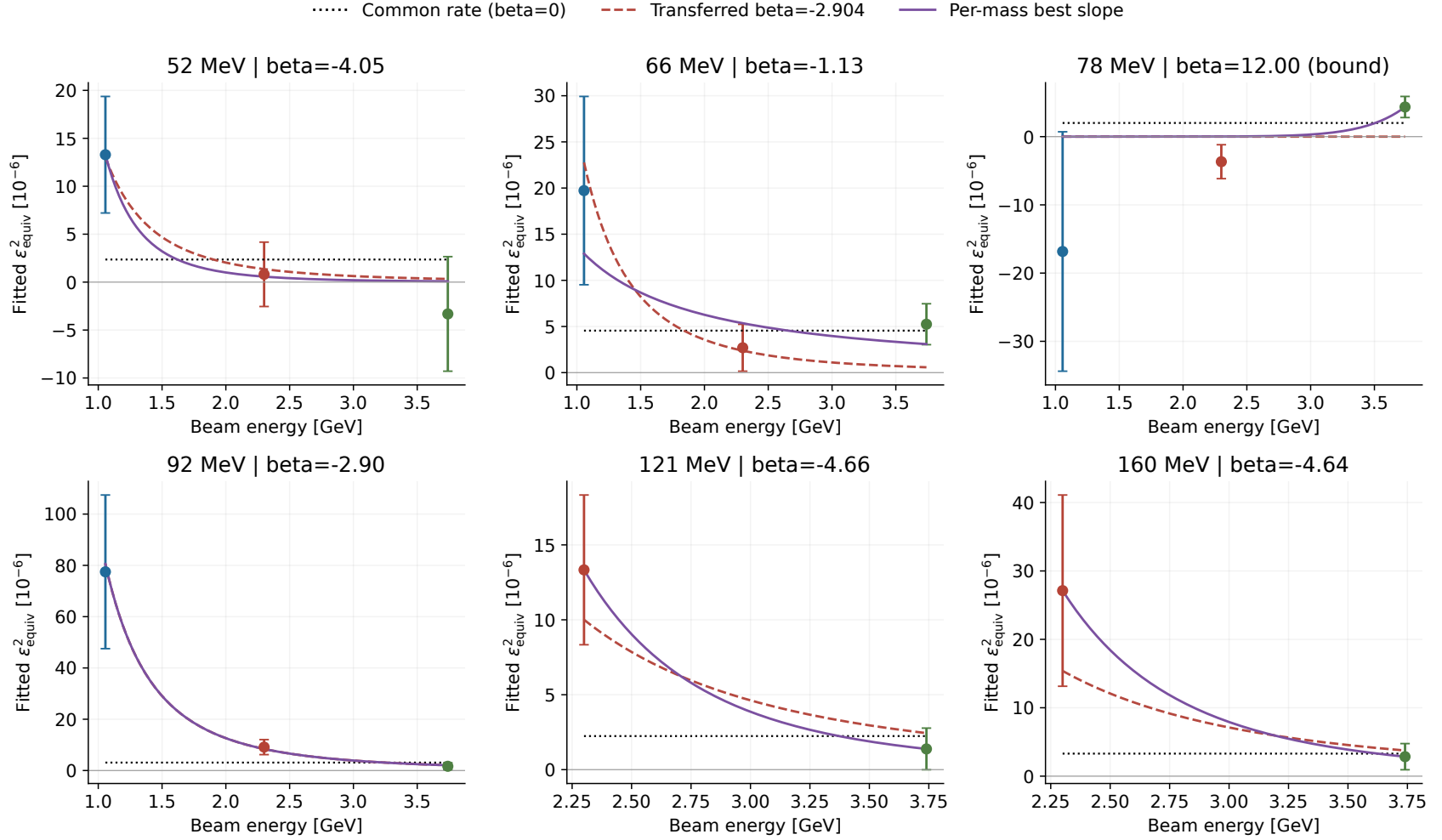


Figure 2: Standalone signed amplitudes and local curvature errors, compared with a common rate, the transferred 92 MeV slope, and the per-mass best slope. Colors of points identify 2015 (blue), 2016 (red) and 2021 (green). Curve amplitudes are separately refitted. All curves predict nonnegative signal; negative data points describe deficits. Only 2016+2021 are available at 121 and 160 MeV. The 52 MeV fit is dominated by 2015 and is weakly constrained toward very negative slopes; it is not a three-positive-rate example.

All three pair combinations

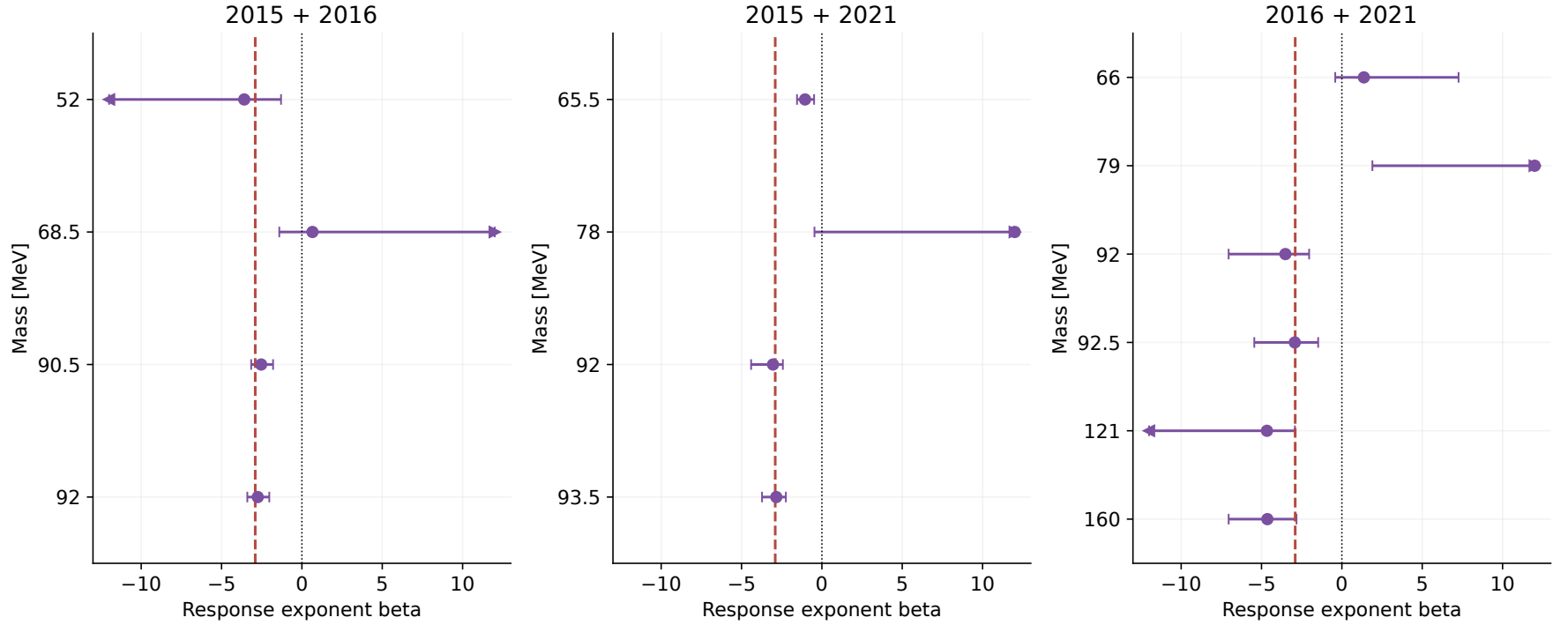


Figure 3: Every retained candidate for each pair, with 92 MeV added as a common reference. Bars are connected unit-deviance profile spans, arrows indicate truncation at ± 12 , and the red line is the frozen -2.904 slope. A boundary point is an optimizer result under a finite search range, not a measured exponent. If both separate pair amplitudes $\hat{a}_i$ and $\hat{a}_j$ are strictly positive, then

$$\hat{\beta}_{ij} = \frac{\log(\hat{a}_i/\hat{a}_j)}{\log(E_i/E_j)}, \quad \hat{\mu} = \hat{a}_i (E_i/2.30 \text{ GeV})^{-\hat{\beta}_{ij}}.$$

When this slope lies inside the allowed range, it exactly reproduces both separate best-fit rates. All 23 applicable pairs pass this saturation check to numerical precision. If one independent physical amplitude is zero, exact saturation can require an infinite slope; if both vanish, the slope is unidentified.

The 2015+2021 candidate at 65.5 MeV prefers $\beta = -1.048$; the 2016+2021 candidate at 66 MeV prefers $+1.370$. The selected pair candidates at 90.5, 92.5 and 93.5 MeV lie near the original negative slope; other pair combinations in this region can be unstable. At 121 and 160 MeV, the latter pair prefers -4.662 and -4.635 , but the transferred slope costs only 1.022 and 0.927 in twice NLL. These are broad, descriptive comparisons. The complete table follows; none is assigned a compatibility probability or counted as an independent discovery test.

What a pair predicts for the unused dataset at 92 MeV

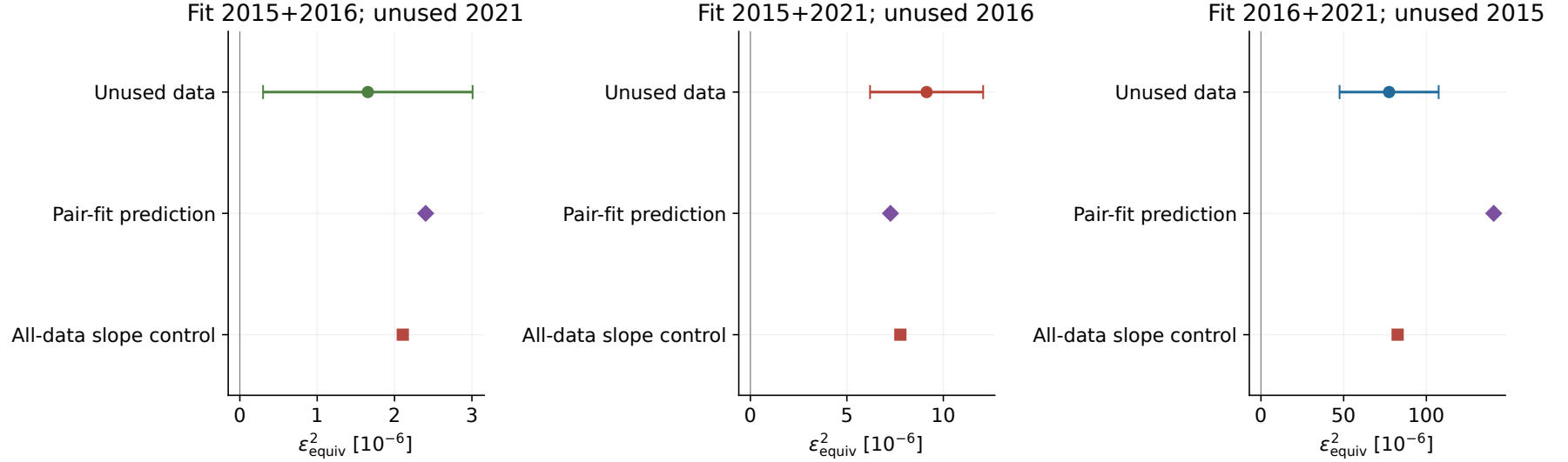


Figure 4: Purple points freeze the amplitude and slope fitted using only the named pair and predict the unused dataset. Their uncertainty is not propagated; bars belong to the unused standalone fit. Red squares are controls using the original all-data 92 MeV slope with the pair amplitude refitted. Because that slope already used all three datasets, red squares are not independent predictions. Mass selection also used the full data.

Training pair	Unused	Prediction	Unused fit	Plug-in D
2015 + 2016	2021	2.40	1.65 ± 1.35	0.305
2015 + 2021	2016	7.26	9.13 ± 2.93	0.408
2016 + 2021	2015	140.74	77.48 ± 29.96	4.450

All amplitudes in the table are in units of 10^{-6} . The plug-in penalty holds the pair prediction fixed and profiles only the unused dataset's background. The 2016+2021 pair extrapolates to 140.74×10^{-6} in 2015, compared with its standalone 77.48 ± 29.96 . Its plug-in penalty is 4.450. Refitting the shared amplitude and slope using all three datasets instead gives a compatibility penalty of 0.157. This refit uses the third dataset and is not a held-out prediction; the difference exposes uncertainty and extrapolation sensitivity in the pair fit.

The full ledger retains 39 unused-dataset comparisons, including unstable cases. Four free-slope extrapolations yield nonpositive profiled backgrounds in the inherited Gaussian constraint model and are flagged invalid for physical interpretation: the 2016+2021 fits at 52, 68.5, 90 and 90.5 MeV extrapolated to 2015. Their numerical objective values are retained only for audit. All 92 MeV comparisons shown here have positive profiled backgrounds.

Complete retained pair-candidate results

Pair	Mass	$\hat{\beta}$	D_0	D_{92}	D_{best}
2015 + 2016	52	-3.59	3.23	0.03	0.000
2015 + 2016	68.5	0.66	0.04	3.67	0.000
2015 + 2016	90.5	-2.53	4.44	0.35	0.000
2015 + 2021	65.5	-1.05	2.31	5.17	0.000
2015 + 2021	78	12.00*	0.52	7.89	0.000
2015 + 2021	93.5	-2.84	5.13	0.01	0.000
2016 + 2021	66	1.37	0.57	3.97	0.000
2016 + 2021	79	12.00*	3.00	6.63	0.005
2016 + 2021	92.5	-2.93	3.47	0.00	0.000
2016 + 2021	121	-4.66	5.31	1.02	0.000
2016 + 2021	160	-4.64	2.96	0.93	0.000

Masses are in MeV. D_0 , D_{92} and D_{best} denote penalties for a common rate, the transferred slope and a free slope. An asterisk marks a slope bound. Zero penalty for a positive pair follows from saturation, not evidence for the power law. All 56 comparisons, including non-candidate pair combinations and controls, are in `derived/response_fits.csv`; complete profiles and unused-data diagnostics are also retained.

Validation, limits and next use

Independent agents checked candidate selection and the physics/statistical interpretation. The computation reproduces the parent’s standalone signed roots to 3.6×10^{-12} and its 92 MeV response fit. All reported common-rate, transferred-rate and best-rate fits retain positive background and total expectations; the four invalid fixed extrapolations are separately marked. Of the 56 comparisons, 7 have zero amplitude and unidentified slope, 17 reach a wide slope bound, and 32 have at least one truncated profile endpoint. These failures of identifiability are part of the result.

The data support further investigation of the local 90–94 MeV response pattern, with 121/160 MeV as broad two-dataset comparisons. They do not establish a common energy law across the spectrum. A physical explanation must predict a residual response after the existing normalization and be checked against detector, target, kinematic and run-period controls. The 2015 extension above 90 MeV, inherited 2016 background qualifications, retrospective selection and correlated mass windows remain in force. The underlying v5.0.4 and v5.5.0 packages are unchanged.

Run `scripts/build.sh` for the fits, five vector figures and note. The package contains input/solver snapshots, selection and fit ledgers, source hashes, numerical checks, rendered-page QA and a SHA-256 manifest. No probabilities, toy ensembles or pooled mass likelihoods were generated.

References

- [1] N. Baltzell et al., *The Heavy Photon Search Experiment*. [arXiv:2203.08324](#), Table I.
- [2] P. H. Adrian et al., *Searching for prompt and long-lived dark photons in electroproduced e^+e^- pairs with HPS*. Phys. Rev. D 108, 012015 (2023). [arXiv:2212.10629](#), Eq. (7).
- [3] J. D. Bjorken, R. Essig, P. Schuster and N. Toro, *New Fixed-Target Experiments to Search for Dark Gauge Forces*. Phys. Rev. D 80, 075018 (2009). [arXiv:0906.0580](#).